UMM AL-QURA UNIVERSITY

College of Engineering & Islamic Architecture

Python-Based Modeling and Simulation of Diclofenac Removal in a Granular Activated Carbon Fixed-Bed Column

Environmental Engineering Graduation Project

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Abstract

Diclofenac is a widely used non-steroidal anti-inflammatory drug that has been detected in water and wastewater, while conventional treatment may not provide reliable removal. Granular activated carbon (GAC) adsorption is a practical polishing process, but fixed-bed performance changes with influent concentration, carbon mass or bed depth, and flow rate. This project develops a Python-based computational framework to predict diclofenac breakthrough behavior and removal efficiency under different operating conditions without claiming that a new laboratory experiment has been performed. Breakthrough-curve data will be extracted from peer-reviewed fixed-bed studies, quality-checked, and represented as the normalized effluent concentration C_t/C_0. Established column models, including Thomas, Yoon-Nelson, Bohart-Adams, and Yan, will be fitted by nonlinear regression and compared using condition-based cross-validation. The model will estimate time-dependent removal efficiency, breakthrough and exhaustion times, total mass adsorbed, and adsorption capacity. A simple Python interface will support scenario testing only within the range covered by the calibration data and will warn against unsupported extrapolation. The expected outcome is a transparent, reproducible decision-support tool that connects adsorption theory, published experimental evidence, and computational simulation while clearly communicating uncertainty and model limitations.

Keywords: *Diclofenac; granular activated carbon; fixed-bed adsorption; breakthrough curve; Python; predictive modeling.*

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1. Introduction

Water quality protection increasingly requires attention to contaminants that are not always removed reliably by conventional treatment. Pharmaceuticals are a prominent group of contaminants of emerging concern because they are continuously introduced into wastewater through household use, hospitals, healthcare facilities, pharmaceutical manufacturing, and the disposal of unused medicines. After administration, a fraction of a medicine may be excreted unchanged or as active metabolites and reach sewer systems. Conventional wastewater-treatment plants are designed mainly to reduce suspended solids, biodegradable organic matter, nutrients, and pathogens; consequently, their performance for trace organic contaminants can vary widely. Recent reviews therefore emphasize the need for targeted tertiary treatment and careful evaluation of process conditions rather than assuming that a single technology will achieve the same removal in every water matrix (Alessandretti et al., 2021).

Diclofenac is a widely used non-steroidal anti-inflammatory drug prescribed for pain and inflammation. Its physicochemical behavior, extensive use, and incomplete removal in some treatment systems have made it an important model compound in water-treatment research. The significance of diclofenac is not limited to its presence alone; persistent low-level discharge can create continuous exposure in receiving waters. Treatment evaluation must therefore consider both the percentage removed at a particular moment and the duration for which the process maintains an acceptable effluent concentration. This distinction is especially important for adsorption columns, whose performance changes over time as available adsorption sites become occupied.

Several treatment options have been investigated for diclofenac, including advanced oxidation, membrane separation, biological processes, and adsorption. Adsorption using activated carbon remains attractive because it is comparatively simple to operate, can be applied as a polishing step, and does not intentionally transform the contaminant into potentially unknown oxidation products. Activated carbon contains a network of pores and surface functional groups that can interact with organic molecules. However, removal depends on the carbon source and activation method, pore-size distribution, surface chemistry, solution pH, temperature, ionic strength, competing solutes, and contaminant concentration. Recent experimental work confirms that activated-carbon properties and operating conditions strongly affect the apparent diclofenac capacity, so a value obtained for one material or condition cannot be transferred automatically to another (Mirzaee et al., 2021; de Azevedo et al., 2024).

Granular activated carbon can be operated in a fixed-bed column in which contaminated water flows continuously through a packed carbon bed. At the beginning of operation, most diclofenac is retained and the effluent concentration is low. As the bed receives more contaminant, adsorption sites near the inlet become progressively occupied and a mass-transfer zone moves through the column. The effluent concentration then rises until the bed approaches exhaustion. This dynamic response is represented by a breakthrough curve, usually plotted as the normalized effluent concentration C_t/C_0 against time or treated volume. Characteristic points on the curve, such as breakthrough, 50% breakthrough, and operational exhaustion, describe the useful service life of the bed more meaningfully than a single removal value.

The shape and position of a breakthrough curve respond to several controllable variables. A larger carbon mass or deeper bed generally increases the available adsorption inventory and may delay breakthrough. A higher flow rate normally reduces hydraulic contact time and can cause earlier breakthrough, although the observed response also depends on dispersion and mass-transfer resistance. Increasing the inlet concentration raises the contaminant loading rate and the concentration driving force simultaneously, so its effect should be quantified from data rather than reduced to a universal rule. Fixed-bed experiments by de Franco et al. (2018) and Fadzail et al. (2022) illustrate how these variables alter column behavior and why operating conditions must accompany every reported performance value.

Laboratory column experiments are essential, but they require equipment, analytical measurements, time, and repeated runs. Computational modeling complements experiments by converting breakthrough data into equations that can estimate service times, compare operating conditions, test sensitivity, and identify scenarios that deserve further experimental investigation. Simple analytical models such as Thomas, Yoon-Nelson, Bohart-Adams, and Yan are commonly used to describe sigmoidal breakthrough curves. Their fitted parameters can be useful for prediction, but they should be interpreted cautiously because good curve fit does not necessarily prove that the model assumptions or parameter meanings are physically correct (Chu, 2023). Validation against complete operating conditions is therefore more informative than fitting and testing on randomly mixed points from the same curve.

Python provides a suitable environment for this work because data processing, nonlinear regression, validation, visualization, uncertainty analysis, and interface development can be combined in one reproducible workflow. Recent machine-learning research on granular activated-carbon adsorbers also demonstrates that computational models can help predict early breakthrough when adequate and well-described training data are available (Koyama et al., 2024). Nevertheless, the present project prioritizes transparent fixed-bed equations and uses more flexible surrogate methods only when the amount of independent data is sufficient. The purpose is not to produce a visually impressive calculator with unsupported precision, but to build a model whose inputs, source data, assumptions, errors, and range of applicability are explicit.

2. Problem Statement

The original project concept used a single adsorption-capacity constant to estimate removal from carbon mass. Although useful as a first calculator, a fixed constant cannot represent the time-dependent behavior of a flow-through column and cannot properly account for changes in influent concentration, flow rate, carbon mass, or breakthrough stage. It may therefore produce apparently precise results that are not supported by the operating conditions.

A more defensible approach is to build a Python model from published breakthrough curves and to fit established fixed-bed equations. The central problem is thus the absence of a validated, transparent computational tool that predicts diclofenac removal performance across the operating conditions represented in the available literature data.

> **Core Research Gap:** *Published fixed-bed studies provide breakthrough data, but the project needs an integrated Python workflow that converts those data into validated predictions, compares alternative models, and reports uncertainty and applicability limits.*

3. Literature Review

3.1 Diclofenac as an Aquatic Contaminant

Recent literature treats diclofenac as an important pharmaceutical micropollutant because its occurrence is linked to continuous use and incomplete removal in some conventional treatment systems. Alessandretti et al. (2021) reviewed diclofenac detection, properties, and tertiary-treatment options and showed that removal performance varies with technology and water characteristics. This variability is central to the present project: a model should not predict a universal removal percentage, but should relate the expected response to a defined adsorbent, influent concentration, flow condition, and time. The reviewed evidence also supports adsorption as a relevant polishing process while recognizing that adsorption transfers diclofenac to a solid phase that must eventually be regenerated or disposed of responsibly.

3.2 GAC Adsorption and Breakthrough Behavior

Adsorption transfers dissolved diclofenac from water to external surfaces and internal pores of activated carbon. The process may involve film transfer, pore diffusion, surface interactions, and competition for adsorption sites. Mirzaee et al. (2021) studied diclofenac adsorption on mesoporous activated carbons produced by physical and chemical activation and combined experiments with genetic-programming and molecular-dynamics approaches. Their work demonstrates two points relevant to DicloPure: carbon preparation strongly influences performance, and computational methods can reveal relationships that are difficult to summarize with one fixed capacity. De Azevedo et al. (2024) likewise combined experimental uptake of potassium diclofenac with modeling, reinforcing the current direction toward data-informed prediction rather than isolated percentage-removal claims.

In continuous fixed-bed operation, equilibrium capacity alone is insufficient because the effluent changes with time. Removal depends on bed inventory, hydraulic residence time, the concentration driving force, mass-transfer resistance, axial dispersion, and the progression of the mass-transfer zone. A breakthrough curve begins near C_t/C_0 = 0 when the effluent is nearly contaminant-free and approaches C_t/C_0 = 1 as the bed becomes exhausted. The area above the curve can be used in a mass balance to estimate the contaminant retained by the bed. This representation allows a model to answer practical questions about service time, usable capacity, and the effect of changing operating conditions.

3.3 Evidence from Diclofenac Fixed-Bed Studies

De Franco et al. (2018) provide the most directly relevant fixed-bed evidence for the proposed model. Their factorial column study varied inlet diclofenac concentration between 20 and 100 mg/L, activated-carbon mass between 0.5 and 1.0 g, and flow rate between 3 and 5 mL/min. They fitted Thomas, Bohart-Adams, and Yan equations. The Yan equation produced the highest average coefficient of determination for the breakthrough curves, whereas the Thomas equation better predicted the amount adsorbed by the packed bed. The difference between curve-shape accuracy and capacity accuracy is important: model selection should consider multiple outputs, including held-out curve error, characteristic times, and mass-balance capacity, rather than relying on R^2 alone.

Fadzail et al. (2022) evaluated diclofenac sodium removal in a fixed-bed column using activated carbon prepared from Dillenia indica peels. The study varied bed height, inlet concentration, and flow rate and compared Thomas, Yoon-Nelson, and Bohart-Adams descriptions. Although the adsorbent differs from commercial GAC, the study is useful for evaluating whether the same model structures describe another diclofenac-carbon system. It should therefore serve as external model-structure evidence rather than be pooled uncritically with a calibration dataset based on a different carbon. This separation avoids attributing differences in surface chemistry or pore structure to flow or concentration alone.

3.4 Computational Modeling Approaches

Fixed-bed models range from compact analytical expressions to detailed transport equations. The Thomas equation is often used for capacity-oriented fitting; Yoon-Nelson expresses a sigmoid response through a rate parameter and the 50% breakthrough time; Bohart-Adams is commonly applied to the early breakthrough region; and Yan modifies the dose-response form to improve representation across the curve. These labels should not be confused with proof of a unique mechanism. Chu (2023) reviewed common analytical fixed-bed models and warned that inappropriate linearization, misinterpreted parameters, and favorable R^2 values can create misleading conclusions. Accordingly, DicloPure will fit untransformed concentration ratios by nonlinear regression, check residuals and physical parameter bounds, and compare prediction on entire operating conditions withheld from fitting.

3.5 Python, Machine Learning, and the Research Gap

Python supports nonlinear optimization, reproducible data cleaning, condition-based cross-validation, graphical diagnostics, and deployment of a lightweight interface. Koyama et al. (2024) used machine learning to predict early breakthrough of recalcitrant organic micropollutants in GAC adsorbers, demonstrating the value of data-driven treatment prediction and the need for broad training data. Because the expected diclofenac dataset is smaller, DicloPure will establish transparent analytical models first and explore a surrogate model only if the number of independent conditions is adequate. This leaves a need for a traceable Python workflow tailored to diclofenac column data.

4. Significance of the Study

Environmental significance: The project supports evaluation of a treatment option for a pharmaceutical contaminant and helps identify conditions that delay breakthrough and sustain removal performance.

Engineering significance: A validated model can reduce the number of trial calculations required to screen operating scenarios, prioritize experiments, and communicate how column behavior changes with concentration, flow, and carbon inventory.

Scientific significance: The work integrates literature evidence, adsorption theory, numerical fitting, model comparison, uncertainty analysis, and reproducible Python implementation in one transparent framework.

5. Study Objectives

Primary Objective

> **Objective:** *To develop and validate a Python-based computational model that predicts diclofenac breakthrough behavior and removal performance in a GAC fixed-bed column under different initial concentrations, carbon masses, and flow rates.*

Specific Objectives

1. Compile a traceable dataset of diclofenac breakthrough curves and operating conditions from peer-reviewed fixed-bed studies.
2. Calculate time-dependent removal efficiency, breakthrough time, exhaustion time, total adsorbed mass, and bed adsorption capacity from normalized concentration data.
3. Fit and compare the Thomas, Yoon-Nelson, Bohart-Adams, and Yan breakthrough models using nonlinear regression in Python.
4. Quantify the individual and combined effects of initial diclofenac concentration, carbon mass or bed depth, and flow rate on predicted column performance.
5. Validate predictive performance with condition-based cross-validation and report R^2, RMSE, MAE, and errors in characteristic times and adsorption capacity.
6. Create a simple Python interface that plots predicted breakthrough curves, estimates performance indicators, and flags inputs outside the calibrated range.
7. Document uncertainty, data limitations, and the conditions under which model predictions may be used for screening rather than design certification.

6. Research Questions

8. How accurately can a Python-based computational model predict the full diclofenac breakthrough curve, C_t/C_0 versus time, under different operating conditions when evaluated by condition-based cross-validation?
9. Which established fixed-bed model - Thomas, Yoon-Nelson, Bohart-Adams, or Yan - provides the most reliable predictions for the available diclofenac dataset?
10. How do initial diclofenac concentration, carbon mass or bed depth, and flow rate affect predicted removal efficiency, breakthrough time, and exhaustion time?
11. Do interactions among concentration, carbon inventory, and flow rate materially improve prediction compared with single-variable relationships?
12. How accurately can the model estimate the total diclofenac mass removed and the adsorption capacity of the GAC before exhaustion?
13. Within the calibrated data range, which operating combinations are predicted to delay breakthrough and maintain the required removal efficiency for the longest period?

> **Modeling emphasis:** *All six questions now lead back to model construction, validation, sensitivity analysis, and simulation. The original sixth question has become the primary research question.*

7. Proposed Simulation Programs

7.1 Python - Primary Modeling Platform

Python will be the primary platform because it supports reproducible data processing, nonlinear optimization, statistical evaluation, plotting, and interface development. pandas will organize the breakthrough dataset; NumPy will support numerical calculations; SciPy will fit nonlinear model parameters; scikit-learn will provide validation utilities and optional surrogate models; Matplotlib or Plotly will generate curves; and Streamlit may be used for a simple interactive tool.

DicloPure Python Website Code

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt

ADSORPTION_CAPACITY = 12.43

st.title("DicloPure")

carbon_mass = st.number_input("Activated Carbon Mass (g)", value=10.0)
diclofenac_concentration = st.number_input("Diclofenac Concentration (mg/L)", value=20.0)
water_volume = st.number_input("Water Volume (L)", value=5.0)
flow_rate = st.number_input("Flow Rate (mL/min)", value=3.0)

total_diclofenac = diclofenac_concentration * water_volume
removal_capacity = ADSORPTION_CAPACITY * carbon_mass

if removal_capacity >= total_diclofenac:
    removed = total_diclofenac
    efficiency = 100
else:
    removed = removal_capacity
    efficiency = (removed / total_diclofenac) * 100

st.subheader("Results")

st.write(f"Removed Diclofenac: {removed:.2f} mg")
st.write(f"Removal Efficiency: {efficiency:.2f}%")

st.latex(r'''
Diclofenac Removed = 12.43 × Mass of Activated Carbon (g)
''')

data = pd.DataFrame({
    "Category": ["Total", "Removed"],
    "Amount": [total_diclofenac, removed]
})

fig, ax = plt.subplots()
ax.bar(data["Category"], data["Amount"])
st.pyplot(fig)
```

Figure 1. Original DicloPure Python/Streamlit prototype retained from the team presentation.

The prototype establishes the intended interface direction. In the revised implementation, the fixed adsorption-capacity value will be replaced by fitted, condition-specific breakthrough predictions and the interface will report its calibration limits.

7.2 Spreadsheet Software - Verification Only

Excel may be used to inspect extracted values and independently verify selected mass-balance calculations. It will not be the authoritative modeling environment because manual formulas and duplicated worksheets are harder to audit and reproduce than a versioned Python workflow.

7.3 Curve Digitization Tool

When a paper provides breakthrough curves without downloadable numerical values, WebPlotDigitizer or an equivalent tool will be used to recover time and C_t/C_0 points. Each extracted series will retain a source identifier, figure number, operating conditions, and a note that the values were digitized rather than supplied directly by the authors.

7.4 Model Candidates and Their Roles

Model	Primary role	Main output	Important limitation
Thomas	Capacity-oriented empirical fit	q_0 and rate constant	Simplified equilibrium and kinetic assumptions
Yoon-Nelson	Compact prediction of sigmoid breakthrough	50% breakthrough time and rate	Does not explicitly represent every transport mechanism
Bohart-Adams	Early breakthrough-region description	Kinetic and bed-capacity terms	Often less accurate near full saturation
Yan	Modified dose-response fit across the curve	Curve shape and characteristic times	Parameters remain empirical and material-specific
Optional surrogate	Interpolate fitted parameters across conditions	Predicted curve at new in-range inputs	Requires enough independent conditions to avoid overfitting

Table 1. Proposed fixed-bed models and intended use in DicloPure.

> **Integrated strategy:** *Published data -> quality control -> model fitting -> condition-based validation -> sensitivity analysis -> in-range scenario simulation -> transparent Python interface.*

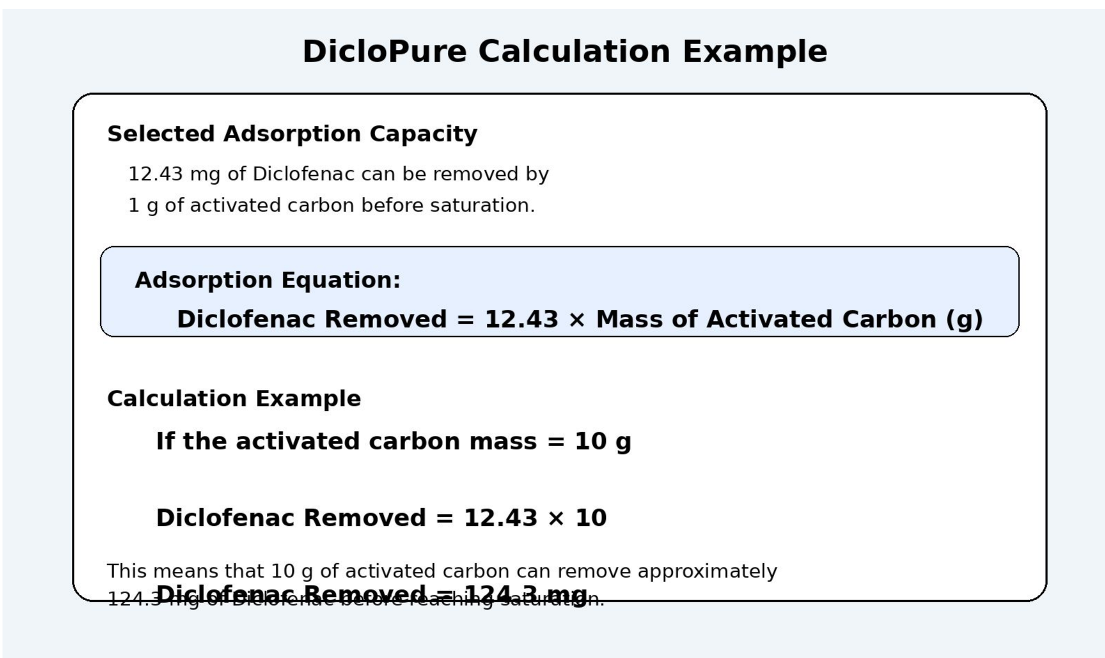

Figure 2. Original constant-capacity DicloPure calculation concept. It is preserved as project evidence but superseded in the final workflow by condition-specific model fitting.

8. Data Required for the Project

Data group	Required fields	Units / form	Purpose

Data group	Required fields	Units / form	Purpose
Breakthrough series	Time, inlet C_0, outlet C_t or C_t/C_0	min; mg/L; dimensionless	Fit curves and derive characteristic times
Operating conditions	Flow rate, carbon mass, bed depth, column diameter	mL/min; g; cm; cm	Explain differences among curves
Water conditions	pH, temperature, ionic strength, competing solutes	reported study units	Assess transferability and confounding
Adsorbent properties	Carbon source, particle size, surface area, pore data	study-specific	Prevent inappropriate pooling of dissimilar carbons
Source metadata	Authors, year, DOI, figure/table, extraction method	text	Maintain traceability and reproducibility

Table 2. Minimum data fields for a reproducible modeling dataset.

The core calibration dataset should use one activated-carbon material and one experimental study whenever possible. Data from different carbons should be analyzed as external comparison or encoded with material descriptors; otherwise, differences in surface chemistry and pore structure may be incorrectly attributed to concentration or flow rate.

9. Data from Previous Projects

Study	Relevant conditions	Modeling contribution	Planned use
de Franco et al. (2018)	C_0: 20 and 100 mg/L; carbon: 0.5 and 1.0 g; Q: 3 and 5 mL/min	Full-factorial fixed-bed curves; Thomas, Bohart-Adams, and Yan fits	Primary calibration candidate
Mirzaee et al. (2021)	Mesoporous activated carbons prepared by physical and chemical activation	Experimental adsorption, genetic programming, and molecular simulation	Material and computational comparison
Fadzail et al. (2022)	Dillenia indica carbon; varied bed height, C_0, and flow	Thomas, Yoon-Nelson, and Bohart-Adams comparison	External model-structure check
Koyama et al. (2024)	GAC adsorbers and recalcitrant organic micropollutants	Machine-learning prediction of early breakthrough	Validation and data-driven modeling benchmark
de Azevedo et al. (2024)	Activated-carbon uptake of potassium diclofenac	Experimental adsorption and modeling	Recent external comparison

Table 3. Previous studies and their role in the proposed project.

> **Data integrity rule:** *Any concentration-ratio table used in the analysis must be traceable to a published table, supplementary file, or documented curve-digitization step.*

Carbon 0.5 g | Flow 3 mL/min

Time (min)	C/C_0
0	0.0
5	0.1
10	0.35
15	0.62
20	0.8
30	0.9
40	0.94
60	0.96
80	0.97
100	0.98
120	0.99

Carbon 1.5 g | Flow 3 mL/min

Time (min)	C/C_0
0	0.0
5	0.06
10	0.22
15	0.45
20	0.68
30	0.84
40	0.91
60	0.94
80	0.95
100	0.96
120	0.97

Carbon 0.5 g | Flow 5 mL/min

Time (min)	C/C_0
0	0.0
5	0.18
10	0.52
15	0.77
20	0.89
30	0.94
40	0.96
60	0.97
80	0.98
100	0.99
120	0.99

Carbon 1.5 g | Flow 5 mL/min

Time (min)	C/C_0
0	0.0
5	0.12
10	0.4
15	0.66
20	0.82
30	0.9
40	0.93
60	0.95
80	0.96
100	0.97
120	0.98

Figure 3. Original DicloPure breakthrough dataset organized by activated-carbon amount and flow rate.

Conc 20 mg/L | Flow 3 mL/min

Time (min)	C/C_0
0	0.0
5	0.08
10	0.25
15	0.45
20	0.62
30	0.78
40	0.86
60	0.9
80	0.91
120	0.93
160	0.95

Conc 100 mg/L | Flow 3 mL/min

Time (min)	C/C_0
0	0.0
5	0.12
10	0.4
15	0.68
20	0.82
30	0.9
40	0.94
60	0.96
80	0.97
120	0.98
160	0.99

Conc 20 mg/L | Flow 5 mL/min

Time (min)	C/C_0
0	0.0
5	0.2
10	0.55
15	0.78
20	0.88
30	0.93
40	0.95
60	0.96
80	0.97
120	0.98
160	0.99

Conc 100 mg/L | Flow 5 mL/min

Time (min)	C/C_0
0	0.0
5	0.28
10	0.65
15	0.84
20	0.92
30	0.96
40	0.97
60	0.98
80	0.99
120	1.0
160	1.0

Figure 4. Original DicloPure breakthrough dataset organized by inlet concentration and flow rate.

> **Project-data status:** *The original tables are retained unchanged. Before model calibration, every series will be traced to its source or documented digitization step and checked for units, operating conditions, and transcription errors.*

10. Study Methodology

10.1 Study Design

This is a secondary-data computational modeling study. It does not claim new laboratory measurements. The unit of analysis is one complete breakthrough curve produced under a defined combination of influent concentration, carbon mass or bed depth, and flow rate. The main response is $y(t) = C_t/C_0$, bounded between 0 and 1.

10.2 Data Acquisition and Quality Control

14. Select peer-reviewed fixed-bed studies that report diclofenac breakthrough curves and operating conditions.
15. Obtain tabulated data or digitize curves at sufficient resolution, preserving source and figure metadata.
16. Convert units consistently and calculate C_t/C_0 when only absolute concentrations are given.
17. Check monotonicity, duplicates, impossible values, missing conditions, and digitization errors; retain justified non-monotonic points rather than silently deleting them.
18. Separate calibration data from external-comparison data when activated-carbon materials or matrices differ substantially.

10.3 Derived Performance Indicators

Removal efficiency at time t: $$R(t) = 100 * [1 - C_t/C_0]$$

Total adsorbed mass: $$M_{ads} = (Q * C_0 / 1000) * \int [1 - C_t/C_0]\, dt$$

For Q in mL/min, C_0 in mg/L, and t in min, M_{ads} is obtained in mg.

Bed adsorption capacity: $$q_e = M_{ads} / m_{GAC}$$

q_e is reported in mg/g when M_{ads} is in mg and m_{GAC} is in g.

For consistent comparison, breakthrough time t_b will be defined at $C_t/C_0 = 0.05$, matching de Franco et al. (2018). The project will define t_{50} at $C_t/C_0 = 0.50$ and operational exhaustion time t_e at $C_t/C_0 = 0.95$. If a source uses different thresholds, both the source definition and the harmonized definition will be recorded.

10.4 Model Fitting

Each candidate equation will be fitted to untransformed C_t/C_0 values by nonlinear least squares. Parameter bounds will be used to prevent nonphysical negative rate or capacity values. Linearized forms may be shown for comparison but will not be the primary fitting method because transforming the response changes the error structure and can overweight the ends of a breakthrough curve.

Yoon-Nelson form: $$C_t/C_0 = 1 / \{1 + \exp[k_{YN} * (\tau - t)]\}$$

Thomas form: $$C_t/C_0 = 1 / \{1 + \exp[k_{Th} * (q_0 * m_{GAC} / Q - C_0 * t)]\}$$

All quantities will be converted to a consistent unit system before fitting.

The Bohart-Adams model will be evaluated mainly for the early breakthrough region, and the Yan model will be fitted across the full curve. These analytical forms will be treated as phenomenological prediction tools rather than proof of a specific adsorption mechanism, and their parameters will not be overinterpreted (Chu, 2023). A hierarchical strategy will first fit each experimental condition independently, then model how fitted parameters change with concentration, flow, and carbon mass. This makes the influence of operating variables explicit and supports in-range prediction.

10.5 Validation and Model Selection

Validation will be performed by leaving out an entire operating condition, fitting the model to the remaining curves, and predicting the held-out curve. Randomly splitting time points from the same curve would leak condition-specific information and overstate accuracy. Performance will be compared using R^2, root mean squared error (RMSE), mean absolute error (MAE), and absolute errors in t_b, t_{50}, t_e, and q_e. Residual plots and parameter uncertainty will also be inspected.

Criterion	Reason for inclusion	Selection principle
Curve RMSE / MAE	Measures pointwise predictive error	Lower held-out error is better
R^2	Summarizes explained variation	Use with residual checks, not alone
Error in t_b and t_e	Directly relevant to service time	Prefer models accurate at both thresholds
Error in q_e	Tests mass-balance usefulness	Avoid a curve fit that misstates capacity
Parameter stability	Supports interpretation and transfer	Reject unstable or nonphysical fits

Table 4. Model-evaluation criteria.

10.6 Simulation and Python Interface

19. Accept initial concentration, carbon mass or bed depth, flow rate, and simulation time as inputs.
20. Check each input against the calibration range and display an extrapolation warning when necessary.
21. Predict C_t/C_0 and removal efficiency over time using the selected model.
22. Report t_b, t_{50}, t_e, M_{ads}, q_e, and uncertainty intervals where estimable.

23. Plot the predicted curve together with calibration data and identify the chosen operating condition.
24. Export a results table and preserve the model version, parameter set, and source dataset identifier.

11. Project Timeline

The timetable below is reproduced directly from the original DicloPure project presentation so that the activity sequence and marked months remain exactly aligned with the team plan.

DicloPure Project Timetable

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Literature Review	XXX	XXX										
Previous Research Analysis	XX	XXX	XX									
Extract Figure 5 Data		XX	XXX	XX								
Data Organization			XX	XXX	XX							
Breakthrough Curve Analysis				XX	XXX	XX						
Apply Adsorption Equation					XX	XXX						
Python Programming						XX	XXX	XX				
AI Prediction Model							XX	XXX	XX			
Results & Discussion								XX	XXX	XX		
Final Report Writing									XX	XXX	XXX	
Presentation Preparation										XX	XXX	XXX

Figure 5. Original DicloPure twelve-month project timetable.

12. Expected Results

30. A quality-controlled dataset linking each breakthrough curve to concentration, carbon inventory, flow rate, and source metadata.
31. A quantitative comparison of Thomas, Yoon-Nelson, Bohart-Adams, and Yan models using held-out operating conditions.
32. Predicted breakthrough curves and removal-efficiency profiles for input combinations within the calibrated range.
33. Estimated t_b, t_50, t_e, total diclofenac mass removed, and GAC adsorption capacity for each simulated condition.
34. Sensitivity results showing the direction and relative importance of concentration, carbon mass or bed depth, and flow rate.
35. A Python interface that communicates predictions, uncertainty, data provenance, and extrapolation warnings.

Based on adsorption theory and the referenced studies, greater carbon inventory is expected to delay breakthrough, whereas greater flow rate is expected to shorten contact time and often move breakthrough earlier. Higher influent concentration increases the loading rate and the mass-transfer driving force, so its net effect must be evaluated from the fitted curves rather than reduced to a single unqualified rule.

13. Conclusion

DicloPure is best defined as a Python-based modeling and simulation project for GAC fixed-bed adsorption of diclofenac. The revised design places the computational question at the center: published breakthrough data are converted into validated predictions of removal efficiency, characteristic service times, and adsorption capacity under different operating conditions. Comparing several established equations, validating by complete operating condition, and warning against extrapolation make the approach more defensible than a calculator based on one constant capacity. The resulting tool is expected to support transparent scenario analysis and environmental-engineering learning while remaining explicit that secondary-data modeling does not replace laboratory or pilot-scale validation.

14. References

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